

Chapter 2

Variational Autoencoders

Exam Study Guide, Worked Concepts, Sample Questions, and Flashcards

Prepared from	Uploaded course materials
Main source	Chapter_2_Variational_Auto_Encoders.pdf
Exam source	Chapter_2_Summary_and Sample_Questions.pdf
Practice source	Chapter_2_Flashcards.pdf

How to use it

This guide focuses on the ideas most likely to appear in an exam: why VAEs are needed, the probabilistic model, marginal likelihood, ELBO, Jensen's inequality, reparameterization, sampling, latent-space geometry, disentanglement, β -VAE, CVAE, and failure modes.

Study order: first read the concept sections, then solve the five sample exam questions, then test yourself with the flashcards.

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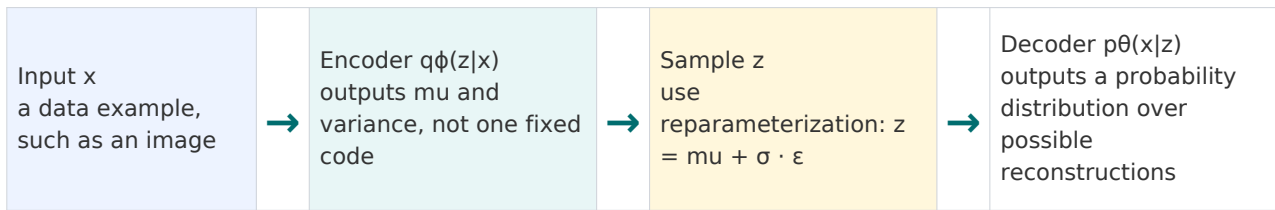
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Most useful mental shortcut

Exam strategy: for every VAE question, identify whether it is asking about (1) the generative model, (2) the inference model, (3) the ELBO objective, or (4) latent-space behavior.

1. Quick exam map

The chapter is about why a standard autoencoder is not enough as a generator, and how a VAE fixes this using a probabilistic latent-variable model.



Exam topic	What you must be able to say
Why VAE?	A deterministic AE reconstructs but does not define a principled probability model $p_\theta(x)$. Its latent space may contain holes.
Generative story	Sample z from $p(z)$, then generate x from $p_\theta(x z)$.
Marginal likelihood	$p_\theta(x) = \int p_\theta(x z)p(z) dz$. This is usually intractable.
Approximate posterior	Use $q_\phi(z x)$ because the true posterior $p_\theta(z x)$ depends on the intractable marginal likelihood.
ELBO	A tractable lower bound on $\log p_\theta(x)$, made of reconstruction likelihood minus KL regularization.
Reparameterization	Moves randomness into ϵ so gradients can pass through z .
Disentanglement	A standard VAE does not guarantee factor separation; β -VAE and CVAE add useful biases.

2. Deterministic Autoencoder vs. Variational Autoencoder

Deterministic Autoencoder

A deterministic autoencoder maps each input to one exact latent code, then reconstructs the input.

$$x \rightarrow z \rightarrow \hat{x}$$
$$\text{encoder: } z = f_{\phi}(x)$$
$$\text{decoder: } \hat{x} = g_{\theta}(z)$$

This is useful for compression, denoising, or reconstruction. The problem is that it does not define a probability model for the data. It does not tell us how likely x is, and it does not give a reliable way to sample new data.

Variational Autoencoder

A VAE changes the encoder from a deterministic mapping to a distribution. Instead of one code z , the encoder outputs parameters of a distribution over z .

$$x \rightarrow (\mu_{\phi}(x), \log \sigma_{\phi}^2(x))$$
$$z \sim q_{\phi}(z|x)$$
$$x \sim p_{\theta}(x|z)$$

Point	Autoencoder	Variational Autoencoder
Encoder output	One fixed code z	A distribution $q_{\phi}(z x)$, usually Gaussian
Decoder output	One reconstruction $\hat{x}$	A likelihood $p_{\theta}(x z)$
Training pressure	Only reconstruct well	Reconstruct well and keep latent space close to a prior
Sampling	Often fails because latent space may have holes	Meaningful because latent codes are regularized toward $p(z)$

Memory line

A good exam sentence: An AE learns a reconstruction function, while a VAE learns a probabilistic latent-variable generative model.

3. The probabilistic model behind a VAE

A VAE assumes that every observed datapoint x was generated from an unobserved latent variable z . The latent variable is the hidden cause, and the decoder explains how that cause produces data.

Generative story

- 1) Sample latent code: $z \sim p(z)$
- 2) Generate observation: $x \sim p_\theta(x|z)$

Standard choices

Component	Meaning	Common choice
Prior $p(z)$	Where latent samples are drawn before seeing data	$N(0, I)$
Decoder $p_\theta(x z)$	Likelihood of data given z	Bernoulli for [0,1] pixels, Gaussian for real-valued data
Encoder $q_\phi(z x)$	Approximate posterior used for inference	Diagonal Gaussian

Joint and marginal distributions

Joint model:

$$p_\theta(x, z) = p_\theta(x|z) p(z)$$

Marginal likelihood:

$$p_\theta(x) = \int p_\theta(x|z) p(z) dz$$

The goal is to learn θ so that the model distribution $p_\theta(x)$ becomes close to the real data distribution $p_{\text{data}}(x)$.

Important

The decoder is stochastic. For a fixed z , it defines a distribution over x . The generated output can vary because we sample z and because the decoder distribution $p_\theta(x|z)$ can also be sampled.

4. Marginal likelihood and why direct optimization is hard

Maximum likelihood learning would maximize $\log p_\theta(x)$. For a latent-variable model this requires integrating over every possible latent explanation z .

$$\log p_\theta(x) = \log \int p_\theta(x|z) p(z) dz$$

This is difficult because the $\int$ is over a high-dimensional continuous latent space, and $p_\theta(x|z)$ is usually produced by a nonlinear neural network. There is normally no closed-form solution.

Why naive Monte Carlo is not enough

A naive idea is to sample many z values from the prior and average $p_\theta(x|z)$. In practice this can be noisy because most random z values may not explain the specific x well. Taking the log of a noisy average makes the estimate unstable, especially with few samples.

Issue	Why it matters in training
High-dimensional z	Many random samples are needed to cover useful regions.

Issue	Why it matters in training
Nonlinear decoder	Each likelihood evaluation is expensive.
Log of average	The estimator can have high variance and biased-looking behavior with small sample counts.
Noisy gradients	Training becomes slow or unstable.

Key transition
This is why the VAE introduces the ELBO: a differentiable, efficient lower bound on $\log p_{\theta}(x)$.

5. Variational inference and approximate posterior

If the true posterior $p_\theta(z|x)$ were available, the model would know which latent values explain a given x . But the true posterior contains $p_\theta(x)$ in its denominator, and $p_\theta(x)$ is the intractable $\int$.

True posterior:

$$p_\theta(z|x) = p_\theta(x|z)p(z) / p_\theta(x)$$

Therefore, the VAE introduces an approximate posterior $q\phi(z|x)$. This is the encoder. It is not assumed to be exactly correct; it is a tractable family we can optimize.

Typical VAE encoder:

$$q\phi(z|x) = N(\mu\phi(x), \text{diag}(\sigma\phi^2(x)))$$

The neural network outputs:

$$\mu\phi(x) \text{ and } \log \sigma\phi^2(x)$$

Distribution	Question it answers
$p(z)$	Before seeing x , where do latent codes usually live?
$p_\theta(x z)$	If z is given, how likely is x ?
$p_\theta(z x)$	After seeing x , which z probably caused it?
$q\phi(z x)$	A tractable approximation to $p_\theta(z x)$ produced by the encoder.

Exam distinction

The encoder is an inference model. The decoder is a generative model. The VAE trains both together.

6. ELBO: formula and meaning

The Evidence Lower Bound (ELBO) is the objective VAEs maximize instead of the intractable log marginal likelihood.

$$\begin{aligned} \mathcal{L}_{\text{ELBO}}(x) &= E_{q\phi(z|x)} [\log p_\theta(x|z)] \\ &- \text{KL}(q\phi(z|x) \parallel p(z)) \end{aligned}$$

Term	Meaning	Effect
$E_{q\phi}[\log p_\theta(x z)]$	Expected log-likelihood / reconstruction term	Encourages z to preserve enough information to reconstruct or explain x .
$\text{KL}(q\phi(z x) \parallel p(z))$	Regularization term	Pushes the approximate posterior toward the prior so the latent space is smooth and sampleable.

Lower-bound identity

$$\log p_\theta(x) = \mathcal{L}_{\text{ELBO}}(x) + \text{KL}(q\phi(z|x) \parallel p_\theta(z|x))$$

Since $\text{KL} \geq 0$:

$$\mathcal{L}_{\text{ELBO}}(x) \leq \log p_\theta(x)$$

The bound becomes exact when the approximate posterior equals the true posterior: $q\phi(z|x) = p_\theta(z|x)$. In that case the KL gap is zero.

7. Jensen's inequality in the ELBO

The ELBO comes from applying Jensen's inequality to the logarithm. The log function is concave, so the log of an expectation is at least the expectation of the log.

For $Y > 0$ and log concave:

$$\log E[Y] \geq E[\log Y]$$

Start with the marginal likelihood and multiply/divide by $q\phi(z|x)$:

$$\begin{aligned} \log p\theta(x) &= \log \int p\theta(x|z)p(z) dz \\ &= \log \int q\phi(z|x) \cdot [p\theta(x|z)p(z)/q\phi(z|x)] dz \\ &= \log E_{q\phi(z|x)} [p\theta(x|z)p(z)/q\phi(z|x)] \end{aligned}$$

Now apply Jensen's inequality:

$$\begin{aligned} \log p\theta(x) &\geq E_q [\log p\theta(x|z) + \log p(z) - \log q\phi(z|x)] \\ &= E_q [\log p\theta(x|z)] - \text{KL}(q\phi(z|x) \parallel p(z)) \end{aligned}$$

Exam explanation

In words: the difficult 'log of an $\int$ /expectation' becomes a tractable 'expectation of a log'. This is why Jensen's inequality is central to the ELBO.

8. Training loss: reconstruction + KL

The theoretical objective maximizes ELBO. Neural-network training usually minimizes a loss, so we minimize negative ELBO.

Maximize:

$$\mathcal{L}_{\text{ELBO}} = \text{reconstruction likelihood} - \text{KL}$$

Minimize:

$$\begin{aligned} \text{Loss} &= -\text{ELBO} \\ &= -E_q [\log p\theta(x|z)] + \text{KL}(q\phi(z|x) \parallel p(z)) \\ &= \text{reconstruction loss} + \text{KL} \end{aligned}$$

Decoder likelihood	Practical reconstruction loss
Bernoulli $p\theta(x z)$	Binary cross-entropy (BCE)
Gaussian $p\theta(x z)$ with fixed variance	Mean squared error (MSE) up to constants

Common confusion

If you see KL subtracted, the paper is maximizing ELBO. If you see KL added, the implementation is minimizing negative ELBO. They are the same objective written with opposite signs.

9. The reparameterization trick

The encoder distribution $q\phi(z|x)$ depends on learnable parameters ϕ . If we sample z directly from this distribution, the random sampling step blocks ordinary backpropagation.

Problematic view:

$$z \sim N(\mu\phi(x), \sigma\phi^2(x))$$

The reparameterization trick rewrites the same sample using a fixed noise source ϵ .

$$\epsilon \sim N(0, I)$$

$$z = \mu\phi(x) + \sigma\phi(x) \cdot \epsilon$$

Fixed noise
 ϵ has no learnable
parameters



Differentiable transform
 $\mu\phi$ and $\sigma\phi$ come from the
encoder



Sample z
 z is now part of the
computation graph

Now gradients can flow through z back to $\mu\phi(x)$, $\sigma\phi(x)$, and the encoder parameters. This gives low-variance gradients and makes VAEs trainable with standard gradient descent.

Why Gaussian latents are common	Reason
Simple sampling	A Gaussian sample can be written as $\mu + \sigma \cdot \epsilon$.
Closed-form KL	KL from a diagonal Gaussian to $N(0, I)$ is easy to compute.
Stable gradients	The reparameterized estimator usually has lower variance than score-function estimators.

One-sentence answer

Conceptually, reparameterization asks: how can I express random sampling as a deterministic transformation of fixed noise?

10. Sampling: why AE fails and VAE works

A standard autoencoder only learns where to place training examples in latent space. It does not force latent codes to match a known distribution. The result can be disconnected clusters with empty holes.

AE sampling attempt:

$$z \sim N(0, I)$$

$$\hat{x} = g\theta(z)$$

Problem: z may fall in a latent-space hole → unrealistic output.

A VAE explicitly regularizes $q\phi(z|x)$ toward $p(z)$, usually $N(0, I)$. This makes random samples from the prior much more likely to land in regions the decoder understands.

VAE sampling pipeline:

$$z \sim p(z) = N(0, I)$$

$$x \sim p\theta(x|z)$$

Main difference

AE: reconstructs well but sampling is ill-defined. VAE: reconstructs and organizes latent space for generation.

11. Latent geometry, entanglement, and disentanglement

A useful VAE latent space has geometry: nearby points tend to decode to similar outputs, interpolation is meaningful, and random samples from the prior decode to plausible data.

Concept	Meaning
Entangled latent representation	A single latent coordinate mixes multiple semantic factors, or one factor is spread across several coordinates.
Disentangled latent representation	Different latent coordinates correspond to different independent semantic factors, such as digit identity, stroke thickness, rotation, or amplitude.
Why it matters	It improves interpretability, controllability, and factor-wise manipulation.

Why a standard VAE does not automatically disentangle

The standard objective asks the model to reconstruct well and keep the posterior near the prior. Those pressures do not force one latent coordinate to represent only one semantic factor. Many equally good parameterizations can explain the same data.

Important limitation

Disentanglement usually requires extra inductive bias, supervision, or structure. It does not appear automatically just because we train a VAE.

12. β -VAE and CVAE

β -VAE

A β -VAE increases the weight of the KL term.

$$\mathcal{L}_{\beta\text{-VAE}}(x) = \mathbb{E}_q[\log p_\theta(x|z)] - \beta \cdot \text{KL}(q\phi(z|x) \parallel p(z))$$

When $\beta > 1$	Effect
Stronger KL pressure	The model is encouraged to use a smaller, more factorized latent code.
Possible benefit	Can encourage disentanglement.
Possible cost	If β is too large, reconstruction quality drops or weaker factors are ignored.

Conditional VAE (CVAE)

A CVAE gives the model known information y , such as a class label, during encoding and decoding.

Encoder: $q\phi(z|x, y)$

Decoder: $p_\theta(x|z, y)$

Conditional ELBO:

$$\mathcal{L}_{\text{ELBO}}(x, y) = \mathbb{E}_{\{q\phi(z|x, y)\}}[\log p_\theta(x|z, y)] - \text{KL}(q\phi(z|x, y) \parallel p(z|y))$$

Conditioning reduces the burden on z . If y already explains a factor, z does not need to encode it. This can reduce entanglement and improve controllability.

13. Common VAE failure modes

Failure mode	What happens	Why it happens	Exam consequence
Posterior collapse	$q\phi(z x)$ becomes almost $p(z)$ for all x .	Decoder too powerful or KL pressure too strong/too early.	z carries little information; decoder ignores the latent code.
Blurry reconstructions	Generated outputs look averaged or smooth.	Common with simple Gaussian likelihood and pixel-wise losses.	Good likelihood objective may still produce visually blurry samples.
Poor sampling	Reconstructions are okay, but samples from $p(z)$ are bad.	Prior and aggregated posterior mismatch or latent space not well organized.	VAE is not functioning well as a generator.
Large ELBO gap	ELBO improves but does not tightly match $\log p\theta(x)$.	Approximate posterior family is too simple, e.g., diagonal Gaussian cannot match true posterior.	Optimizing the bound may not fully improve the true model evidence.
Too much β	Latent variables become too compressed.	KL dominates reconstruction term.	Weak factors may be ignored; reconstructions degrade.

Posterior collapse in one line

$q\phi(z|x) \approx p(z)$ for every $x \Rightarrow z$ contains almost no input-specific information

This is a major failure because the VAE is supposed to learn an informative hidden representation. If the encoder always outputs the prior, the latent variable no longer explains the individual input.

14. One-page formula cheat sheet

Formula	Interpretation
$p(z) = N(0, I)$	Standard latent prior.
$p\theta(x, z) = p\theta(x z)p(z)$	Joint model of data and latent variable.
$p\theta(x) = \int p\theta(x z)p(z) dz$	Marginal likelihood; hard to compute directly.
$q\phi(z x) = N(\mu\phi(x), \text{diag}(\sigma\phi^2(x)))$	Approximate posterior / encoder.
$\mathcal{L}_{\text{ELBO}} = E_q[\log p\theta(x z)] - \text{KL}(q\phi(z x) \ p(z))$	Objective to maximize.
$\log p\theta(x) = \text{ELBO} + \text{KL}(q\phi(z x) \ p\theta(z x))$	ELBO is a lower bound because $\text{KL} \geq 0$.
$\text{Loss} = -E_q[\log p\theta(x z)] + \text{KL}$	Negative ELBO, minimized in practice.
$z = \mu\phi(x) + \sigma\phi(x) \cdot \epsilon$	Reparameterization trick.
$\mathcal{L}_\beta = E_q[\log p\theta(x z)] - \beta \text{KL}$	β -VAE objective.
$q\phi(z x, y), p\theta(x z, y)$	CVAE encoder and decoder condition on y .

15. Sample exam questions from the main summary

These five questions are the main exam-style questions from the uploaded summary, rewritten with clear, complete answers.

Question 1

Why is the ELBO called a lower bound on $\log p_{\theta}(x)$, and when does it become exact?

Sample answer: The ELBO is a lower bound because $\log p_{\theta}(x) = \mathcal{L}_{\text{ELBO}}(x) + \text{KL}(q_{\phi}(z|x) \parallel p_{\theta}(z|x))$. KL divergence is always nonnegative, so $\mathcal{L}_{\text{ELBO}}(x) \leq \log p_{\theta}(x)$. The bound becomes exact when $q_{\phi}(z|x)$ equals the true posterior $p_{\theta}(z|x)$, because the KL gap becomes zero.

Question 2

Interpret the two main terms in the ELBO and explain the tension between them.

Sample answer: The expectation term $E_q[\log p_{\theta}(x|z)]$ rewards the model for explaining or reconstructing x using latent values sampled from the encoder. The KL term pushes $q_{\phi}(z|x)$ toward $p(z)$, organizing the latent space for sampling. The tension is that reconstruction wants z to store detailed input information, while KL regularization wants z to stay simple and close to the prior.

Question 3

Explain the reparameterization trick conceptually. Why is it more than a coding convenience?

Sample answer: The trick rewrites a latent sample as $z = \mu_{\phi}(x) + \sigma_{\phi}(x) \cdot \epsilon$, where $\epsilon \sim N(0, I)$. The randomness is isolated in ϵ , and the learnable parameters appear inside a differentiable transformation. This allows gradients to flow through z into the encoder, making VAE training practical with backpropagation.

Question 4

Why does a standard VAE not automatically guarantee disentangled latent variables? How do β -VAE and CVAE try to change this?

Sample answer: A standard VAE encourages reconstruction and posterior regularity, but it does not force each latent coordinate to represent one semantic factor. β -VAE increases the KL weight, encouraging a more compressed and potentially factorized latent code. CVAE supplies known information y to the encoder and decoder, so z can focus on residual unknown variation. Both add inductive bias, but neither is a perfect guarantee.

Question 5

Suppose $q_{\phi}(z|x)$ is almost identical to $p(z)$ for every input. What does this suggest, and what happens to reconstruction quality?

Sample answer: This suggests posterior collapse. The latent variable carries little or no information about the specific input x , so the decoder effectively ignores z . If the decoder is powerful, it may generate generic-looking outputs, but individual reconstructions usually suffer because the model is not using input-specific latent information.

16. Flashcards for rapid review

Use these for active recall. Cover the answer column and try to answer aloud before checking.

Card 1 Q: What type of mathematical function is the VAE decoder? A: A stochastic/probabilistic function: it defines $p_{\theta}(x z)$, a distribution over x.	Card 2 Q: What is the first source of variability in VAE generation? A: Sampling different latent variables z from the prior $p(z)$.
Card 3 Q: What is the second source of variability in VAE generation? A: Sampling x from the conditional decoder distribution $p_{\theta}(x z)$.	Card 4 Q: Conceptually, what does z map to in the decoder? A: It maps to a probability distribution over the data space X.
Card 5 Q: What is the marginal likelihood formula? A: $p_{\theta}(x) = \int p_{\theta}(x z)p(z) dz$.	Card 6 Q: Why is $\log p_{\theta}(x)$ usually not optimized directly? A: The $\int$ over latent space is usually intractable for neural decoders.
Card 7 Q: What do we maximize instead of $\log p_{\theta}(x)$? A: The Evidence Lower Bound, or ELBO.	Card 8 Q: What does ELBO stand for? A: Evidence Lower Bound.
Card 9 Q: Write the standard ELBO formula. A: $\mathcal{L}_{ELBO} = E_q[\log p_{\theta}(x z)] - KL(q\phi(z x) p(z))$.	Card 10 Q: What does $E_q[\log p_{\theta}(x z)]$ represent? A: Expected reconstruction quality / expected log-likelihood of x.
Card 11 Q: What does $KL(q\phi(z x) p(z))$ represent? A: Regularization that keeps the approximate posterior close to the prior.	Card 12 Q: What is the generative interpretation of maximizing ELBO? A: It pushes $p_{\theta}(x)$ toward the real data distribution while learning useful inference.
Card 13 Q: What does a deterministic AE learn? A: A mapping $x \rightarrow z \rightarrow \hat{x}$ optimized mainly for reconstruction.	Card 14 Q: What does the VAE encoder output? A: Parameters of a distribution, usually $\mu\phi(x)$ and $\log \sigma\phi^2(x)$.
Card 15 Q: What is the standard VAE prior? A: $p(z) = N(0, I)$.	Card 16 Q: What is the decoder likelihood? A: $p_{\theta}(x z)$, the probability model for data conditioned on z.
Card 17 Q: When is a Bernoulli decoder common? A: For normalized binary or [0,1] image pixels; training often uses BCE.	Card 18 Q: When is a Gaussian decoder common? A: For real-valued data; with fixed variance it corresponds to MSE.

16. Flashcards for rapid review - continued

Card 19 Q: What is $q\phi(z x)$? A: The approximate posterior, implemented by the encoder.	Card 20 Q: What is the true posterior formula? A: $p\theta(z x) = p\theta(x z)p(z) / p\theta(x)$.
Card 21 Q: Why do VAEs need variational inference? A: Because the true posterior is intractable, so we optimize a tractable approximation.	Card 22 Q: What is the joint VAE model? A: $p\theta(x,z) = p\theta(x z)p(z)$.
Card 23 Q: Why can AE latent spaces have holes? A: Nothing forces training codes to fill or match a known prior distribution.	Card 24 Q: Why can AE sampling fail? A: Random z values can land in holes where the decoder was never trained.
Card 25 Q: Why is VAE sampling more meaningful? A: KL regularization aligns $q\phi(z x)$ with a known prior $p(z)$.	Card 26 Q: State Jensen's inequality used in ELBO derivation. A: For concave $\log$ and $Y > 0$: $\log E[Y] \geq E[\log Y]$.
Card 27 Q: What is Y in the ELBO derivation? A: $Y(z) = p\theta(x z)p(z)/q\phi(z x)$.	Card 28 Q: When is the ELBO bound tight? A: When $q\phi(z x) = p\theta(z x)$.
Card 29 Q: What is the ELBO gap? A: $KL(q\phi(z x) p\theta(z x))$, the gap between log likelihood and ELBO.	Card 30 Q: Why does the KL sign change in implementations? A: ELBO maximization subtracts KL; loss minimization adds KL.
Card 31 Q: What is the practical VAE loss? A: Loss = reconstruction loss + KL.	Card 32 Q: Write the reparameterization formula. A: $z = \mu\phi(x) + \sigma\phi(x) \cdot \epsilon, \epsilon \sim N(0, I)$.
Card 33 Q: Why does reparameterization help gradients? A: It makes z a differentiable function of encoder outputs and fixed noise.	Card 34 Q: Why are Gaussian latents widely used? A: They have simple reparameterization, closed-form KL to $N(0, I)$, and stable gradients.
Card 35 Q: Why use log variance instead of variance directly? A: It is numerically convenient and helps ensure positive variance after exponentiation.	Card 36 Q: Is latent dimension learned or chosen? A: It is usually a fixed design choice selected before training.
Card 37 Q: What does the KL term encourage geometrically? A: A smooth, organized, prior-aligned latent space.	Card 38 Q: Define entanglement. A: Semantic factors are mixed across latent dimensions.

16. Flashcards for rapid review - continued

Card 39 Q: Define disentanglement. A: Separate latent coordinates correspond to separate semantic factors.	Card 40 Q: Does a standard VAE guarantee disentanglement? A: No. It needs extra inductive bias or supervision.
Card 41 Q: Write the β-VAE objective. A: $\mathcal{L}\beta = \mathbb{E}_q[\log p_\theta(x z)] - \beta \text{KL}(q\phi(z x) \ p(z))$.	Card 42 Q: What happens when $\beta > 1$? A: KL pressure is stronger, encouraging compression and possible factor separation.
Card 43 Q: What is the risk of too large β? A: Poor reconstruction, ignored factors, or posterior collapse.	Card 44 Q: What does a CVAE condition on? A: Known information y, such as class labels or attributes.
Card 45 Q: How does CVAE modify encoder and decoder? A: It uses $q\phi(z x,y)$ and $p_\theta(x z,y)$.	Card 46 Q: How can CVAE reduce entanglement? A: Known factors are handled by y, so z models residual variation.
Card 47 Q: What is posterior collapse? A: $q\phi(z x) \approx p(z)$, so z carries little information about x.	Card 48 Q: Name two causes of posterior collapse. A: An overly powerful decoder and/or KL pressure that is too strong too early.
Card 49 Q: What happens if $q\phi(z x) \approx p(z)$ for all x? A: The encoder stops distinguishing inputs; reconstructions become generic or worse.	Card 50 Q: Why can VAE reconstructions be blurry? A: Simple likelihoods and pixel-wise losses can favor averaged outputs.
Card 51 Q: What is the prior in a VAE? A: The distribution assumed for z before observing data.	Card 52 Q: What is the approximate posterior used for? A: Inferring plausible z values after observing x.
Card 53 Q: Why must prior and posterior roughly agree? A: So samples from the prior land in regions the decoder can generate from.	Card 54 Q: What is the VAE generation pipeline after training? A: Sample $z \sim p(z)$, then sample or decode x from $p_\theta(x z)$.
Card 55 Q: Give a one-sentence summary of a VAE. A: A VAE learns a regularized probabilistic latent space for both inference and generation.	

17. Last-minute exam checklist

Can you answer this?	Expected answer direction
Why is a VAE needed?	Because an AE reconstructs but lacks a principled probability model and sampleable latent prior.
What is $p_{\theta}(x)$?	The marginal likelihood obtained by integrating out z .
Why introduce $q\phi(z x)$?	Because the true posterior is intractable.
Where does Jensen's inequality appear?	It changes $\log E[\dots]$ into $E[\log \dots]$, producing the ELBO.
What are the ELBO terms?	Expected reconstruction likelihood minus KL regularization.
Why does reparameterization matter?	It allows gradients through stochastic latent sampling.
What causes posterior collapse?	z becomes uninformative when $q\phi(z x) \approx p(z)$, often due to strong KL or powerful decoder.
How does β -VAE help?	It strengthens KL to encourage compressed/factorized latents.
How does CVAE help?	It conditions on known y , reducing what z must encode.

Say this aloud before the exam

Final memory paragraph: A VAE is a latent-variable generative model. It samples z from a simple prior, decodes through $p_{\theta}(x|z)$, and trains with the ELBO because $\log p_{\theta}(x)$ is intractable. The ELBO balances reconstruction with KL regularization. Reparameterization makes the stochastic latent sample differentiable. β -VAE and CVAE are ways to add structure to the latent space.

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